

Project Design Phase-II

Data Flow Diagram & User Stories

Date	29 September 2026
Team ID	SWTID-2026-2603
Project Name	AnswerBot – AI-Powered FAQ & Customer Support Automation
Maximum Marks	4 Marks

Data Flow Diagrams

AnswerBot is a secure FAQ management and customer-support automation system. The DFD shows how user requests move through authentication, FAQ search/management, Gemini AI generation, and MongoDB storage.

User Customer Support / Content Creator / Admin	AnswerBot API Node.js + Express.js Authentication + Controllers/Services	MongoDB Users, FAQs, Categories
Login / Search / Create / Generate FAQ	JWT validation → process request → return response	Read / Write FAQ and user data
External AI Service Google Gemini	Flow AI request → generated FAQ draft → user review/edit → save	

Level-0 flow: User → AnswerBot API → Authentication / Business Logic → MongoDB. When AI generation is requested, the API communicates with Google Gemini and returns the generated draft to the user for review.

User Stories

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
Authenticated User	Authentication	USN-1	As a user, I can register with my email and password.	Valid details create an account.	High	Sprint-1
Authenticated User	Authentication	USN-2	As a user, I can log in with valid credentials.	Successful login returns secure authentication.	High	Sprint-1
Authenticated User	FAQ Management	USN-3	As an authorized user, I can create an FAQ.	Required fields are validated and FAQ is saved.	High	Sprint-1
Authenticated User	Search	USN-4	As a user, I can search FAQs using keywords.	Relevant matching FAQs are returned.	High	Sprint-2
Content Creator	FAQ Management	USN-5	As an authorized user, I can edit or delete FAQ content.	Permitted changes are saved successfully.	High	Sprint-2
Content Creator	AI Generation	USN-6	As a user, I can generate an FAQ draft using Google Gemini.	A draft is returned for review/editing.	High	Sprint-3
Content Creator	AI Generation	USN-7	As a user, I can review and edit AI-generated content before saving.	User can modify the draft before persistence.	High	Sprint-3
Administrator	Category Management	USN-8	As an admin, I can manage FAQ categories.	Categories can be created, updated and controlled.	Medium	Sprint-4
Administrator	Security	USN-9	As an admin, I can restrict protected operations by role.	Unauthorized access is rejected.	High	Sprint-4

Team Members

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